

EXPERIMENT 5

Aim

To implement the Support Vector Machine (SVM) algorithm for classification and improve its performance using hyperparameter tuning techniques.

Problem Statement

Heart disease prediction is a binary classification problem where patients must be classified as:

- Presence of heart disease (1)
- Absence of heart disease (0)

The objective is to:

- Build an SVM model for classification
- Apply kernel techniques to handle non-linear data
- Use GridSearchCV for hyperparameter tuning
- Evaluate the model using classification metrics

Dataset Link

Link: <https://www.kaggle.com/datasets>

Methodology

Data Collection

The dataset (**heart.csv**) was loaded using Pandas.

It contains medical attributes such as age, cholesterol level, resting blood pressure, etc., along with a target variable indicating the presence of heart disease.

Data Preprocessing

The dataset was checked for missing or inconsistent values to ensure data quality.

Features (**X**) and target variable (**y**) were separated:

- **X** → input features
- **y** → output label (heart disease: 0 or 1)

This step ensures the dataset is properly structured for model building.

Data Splitting

The dataset was split into:

- **80% Training Data** → used to train the model
- **20% Testing Data** → used to evaluate performance

Stratified sampling was applied to maintain equal class distribution in both sets, reducing bias.

Feature Scaling

Feature scaling was performed using **StandardScaler** to standardize the data:

- Mean = 0
- Standard Deviation = 1

This step is crucial for **SVM (Support Vector Machine)** because it relies on distance and margin calculations, and unscaled features can negatively affect performance.

SVM Model Initialization

An SVM classifier was initialized.

SVM works by finding the optimal hyperplane that separates data points of different classes with maximum margin.

Hyperparameter Tuning

GridSearchCV was used to find the best combination of parameters by performing cross-validation.

Parameters tuned:

- **Kernel** → Determines decision boundary type
 - Linear → straight line
 - RBF → non-linear boundary
- **C (Regularization Parameter)** → Controls trade-off between margin size and classification error
- **Gamma** → Defines influence of individual data points in RBF kernel

This step improves model accuracy and generalization.

Model Training

The SVM model was trained using the best parameters obtained from GridSearchCV on the training dataset.

Prediction

The trained model was used to predict class labels for the test dataset.

Model Evaluation

The model performance was evaluated using:

- **Accuracy Score** → Overall correctness
- **Confusion Matrix** → Displays TP, TN, FP, FN
- **Precision** → Accuracy of positive predictions
- **Recall** → Ability to detect actual positives
- **F1-Score** → Balance between precision and recall

These metrics provide a comprehensive evaluation of the model.

Result

SVM Results

- Achieved high classification accuracy after hyperparameter tuning
- Best parameters selected using GridSearchCV improved performance
- Effectively handled both linear and non-linear decision boundaries

Observations

- Linear kernel works well for simple datasets
- RBF kernel performs better for complex, non-linear data
- Proper tuning of C and gamma is crucial to avoid overfitting

Output:

```
Best Parameters: {'C': 100, 'gamma': 'scale', 'kernel': 'rbf'}

----- SVM Results After Hyperparameter Tuning -----
Accuracy: 1.0

Confusion Matrix:
[[100  0]
 [ 0 105]]

Classification Report:
              precision    recall  f1-score   support

     0           1.00        1.00        1.00        100
     1           1.00        1.00        1.00        105

 accuracy          1.00
 macro avg          1.00
weighted avg          1.00
```

Conclusion:

In this experiment, a Support Vector Machine classifier was implemented on the Heart Disease dataset with hyperparameter tuning using GridSearchCV. Feature scaling improved model performance, and optimal parameters enhanced classification accuracy. The results show that SVM is an effective and reliable algorithm for medical classification tasks.